

Aim - Design and implement a program to simulate a Page Replacement Algorithm using FIFO (First-In-First-Out) in a virtual memory system. The program should load pages into memory frames as needed and replace the oldest page when a page fault occurs. Compute the total number of page faults and page hits for a given reference string, thereby analyzing memory utilization efficiency

```
/c/Users/acom/Desktop/OS_CD24037
GNU nano 8.7
#include <stdio.h>

int main() {
    int frames, n;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter page reference string:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    int frame[frames];
    int front = 0;
    int page_faults = 0, hits = 0;

    // Initialize frames
    for(int i = 0; i < frames; i++) {
        frame[i] = -1;
    }

    for(int i = 0; i < n; i++) {
        int found = 0;

        // Check if page is already in frame
        for(int j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        // If not found → page fault
        if(!found) {
            frame[front] = pages[i];
            front = (front + 1) % frames;
            page_faults++;
        }
    }

    printf("Page Hits = %d\n", hits);
    printf("Page Faults = %d\n", page_faults);

    return 0;
}
```

 /c/Users/acom/Desktop/OS_CD24037

```
acom@DESKTOP-50T9G9Q MSYS ~  
$ cd /c/Users/acom/Desktop/OS_CD24037  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ nano fifo.c  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ gcc fifo.c -o fifo  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ ./fifo  
Enter number of frames: 3  
Enter number of pages: 12  
Enter page reference string:  
1 2 3 4 1 2 5 1 2 3 4 5  
Page Hits = 3  
Page Faults = 9  
  
acom@DESKTOP-50T9G9Q MSYS /c/Users/acom/Desktop/OS_CD24037  
$ |
```